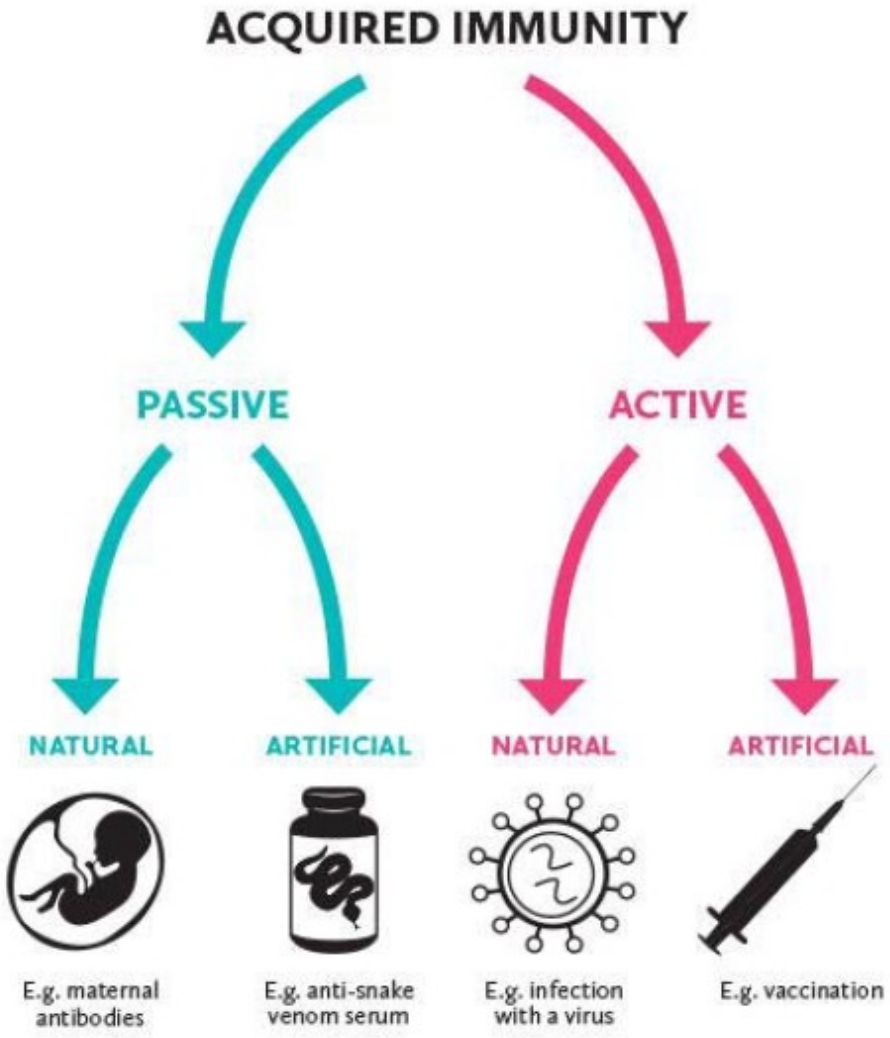
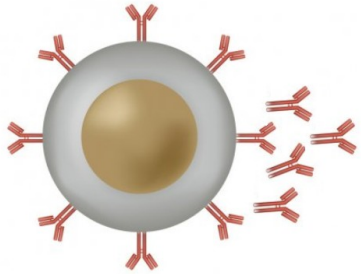
Types of Specific (Acquired) Immunity



Active vs Passive Immunity



ACTIVE

- Exposure to antigen causes body to *actively* produce immunity
  - sensitisation of lymphocytes, producing cell mediated and antibody mediated immunity.
  - memory cells.

*Eg: When a virus naturally enters your body, your body **actively** fights it off via the process above.*



*Eg: When you get a vaccine, your body **actively** produces antibody via the process above*





PASSIVE

- Body gets "ready made" antibody from another source – does not need to actively work to produce the antibody.
  - Antibody received neutralises any pathogens present but will be used up.
  - No memory cells produced

Eg: Antibody produced by the mother passes through the placenta and breast milk to the baby



Eg: Snake antivenom is bottled antibody that is injected into the blood to neutralise venom



Natural vs Artifical Immunity

NATURAL

- Immunity develops via natural processes in the human body.

Eg: When a virus naturally enters your body, your body actively fights it off via cell-mediated and anti-body mediated immunity.



Eg: Antibody naturally produced by the body are passed from mother to baby via the placenta and breast milk so that the baby becomes temporarily, passively immune.



ARTIFICAL

- Scientific processes are used to create immunity.

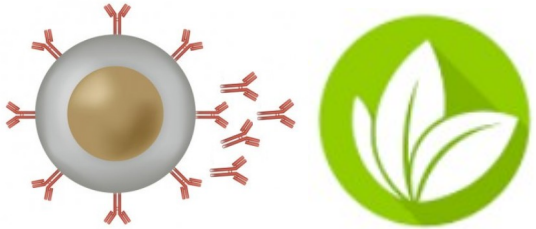
Eg: vaccines are developed to produce active immunity without illness.



Eg: Snake antivenom is produced so that antibodies to the venom can be injected into a person bitten by a snake to neutralise the venom.



Example: Active Natural Immunity



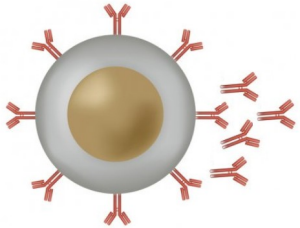
- Eg: Getting the Measles
- If someone is exposed to the Measles virus, their immune system responds.
  - B-cells and T-cells (lymphocytes) become sensitised
  - produce antibody to fight the virus
  - Produce memory cells specific to the virus so next encounter will be neutralised before symptoms develop







Example: Active Artificial Immunity





- Receiving a Measles Vaccine
- A vaccine is given, containing either killed measles virus, substances that mimic the measles antigen, or a very mild form of the virus
  - B-cells become sensitised
  - Antibody produced
  - Memory cells produced
  - Immunity to measles results.

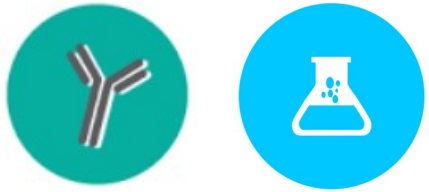


Example: Passive Natural Immunity

- Eg: antibody transfer from mother to child via breastfeeding or across placenta
- Antibodies from mother's bloodstream are transferred to baby during gestation or nursing.
  - Baby then has circulating antibody to neutralise specific pathogen.
  - Doesn't last – no active response from baby's immune system, no memory-cells produced.

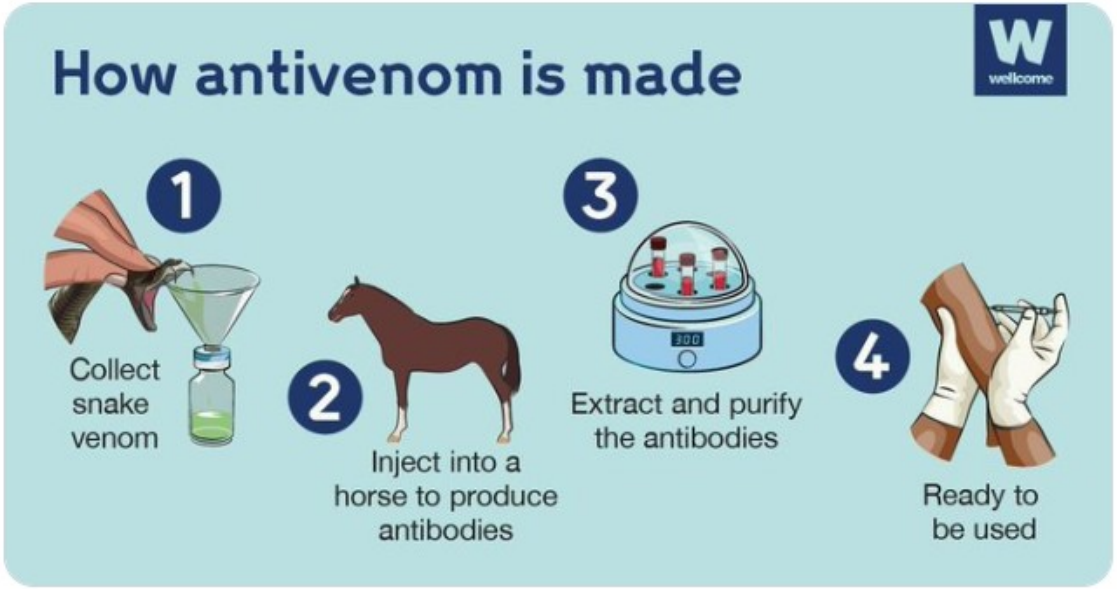


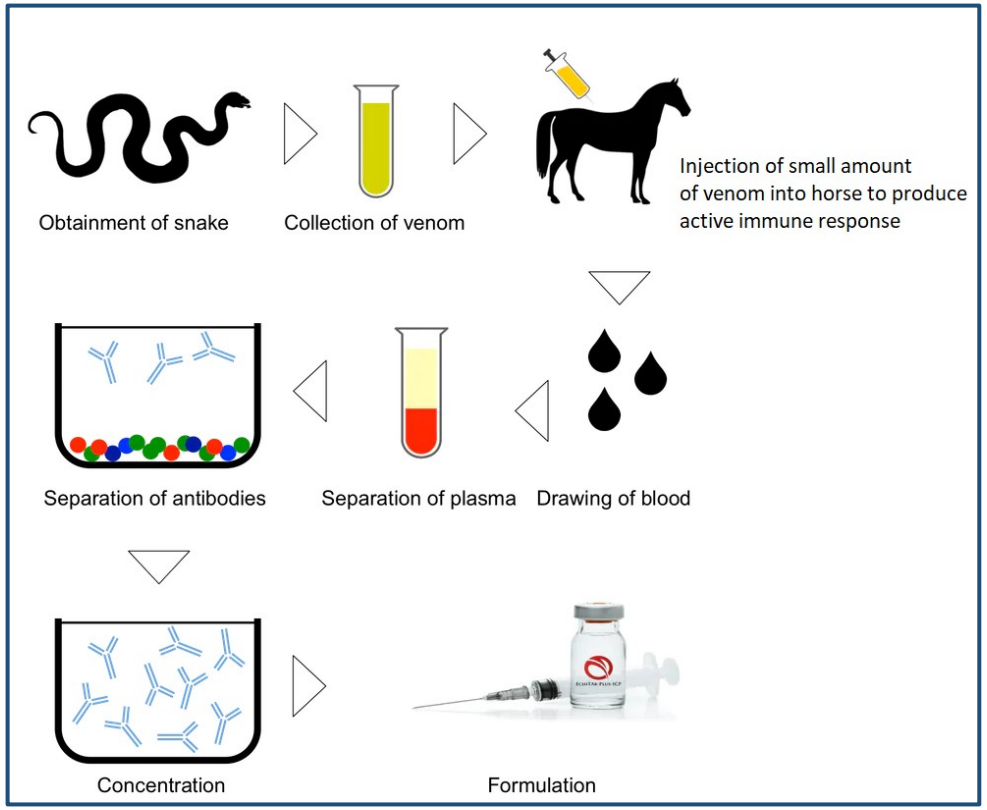
Example: Passive Acquired Immunity

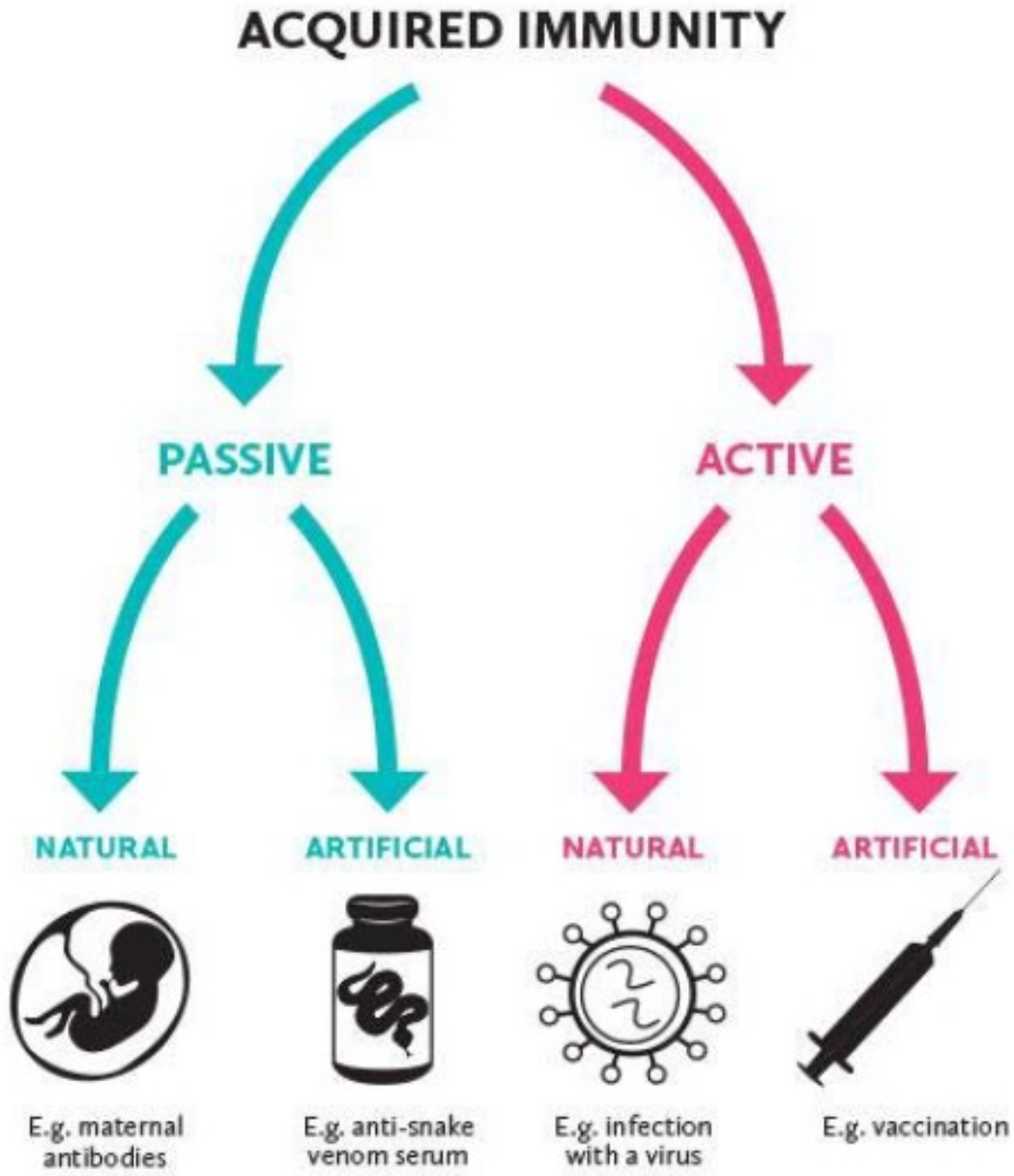




- Eg: administration of snakebite antivenom
- Antibodies to snake venom produced in a laboratory
- Injected into patient's bloodstream:
  - Antibody neutralises venom
  - Short term only – no memory-cells produced






ACQUIRED IMMUNITY
PASSIVE
ACTIVE
NATURAL
ARTIFICIAL
NATURAL
ARTIFICIAL
E.g. maternal antibodies
E.g. anti-snake venom serum
E.g. infection with a virus
E.g. vaccination